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In conjunction with the DeLTA Center at The University of Iowa

"Research in such domains as fetal development, neuroplasticity, the functional organization of the brain, and cognition suggests that the old debates about ***nature and nurture*** *should be thrown out in favor of something new — a unified '****developmental systems****' perspective."*

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Primer



Expertise and individual differences: the search for the structure and acquisition of experts' superior performance

K. Anders Ericsson*

What is expertise and where does it come from? Modern research techniques have made it possible to objectively measure performance in new ways, revealing that expertise derives neither from basic cognitive ability nor from the sheer amount of experience. Rather, it *develops*—particular forms of training and practice induce cognitive, perceptual, physiological, neurological, and anatomical changes necessary for the acquisition of complex domain-specific skills. © 2016 Wiley Periodicals, Inc.

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INTRODUCTION

What is expertise and where does it come from? Modern research techniques have made it possible to objectively measure performance in new ways, revealing that expertise derives not from basic cognitive ability, nor from the sheer amount of experience. Rather, it *develops*—particular forms of training and practice induce cognitive, perceptual, physiological, neurological, and anatomical changes necessary for the acquisition of complex domain-specific skills (see Kaufman and Duckworth, World-class expertise: a developmental model, *WIREs Cogn Sci*, also in the collection How We Develop).

HISTORICAL ORIGINS

The concept of expertise can be traced back to the first cities with their specialized professions, such as shoemakers, blacksmiths, tailors, bakers, and potters. In the Middle Ages, professionals formed city guilds to control the number of individuals and the training process. Within this system, one began as an *apprentice*, took approximately 7 years to become a *journeyman*, and several additional years after that to attain the status of *master*.[1]

During the last two centuries, occupations have become increasingly complex and dynamic, with rapid changes possible in a single career. The old guild system has remained more or less intact in some domains, while others have given way to a system reliant on the recognition of latent abilities—a shift in focus away from development and toward the concept of natural 'aptitudes' for certain domains. According to the aptitude model, intelligence, and aptitude tests are alleged to help measure individual aptitudes so that students can be sorted into their most appropriate professional roles.

OBJECTIVE MEASUREMENT

To evaluate which approach may be more accurate and useful—developmental or aptitude-predictive—we first must be able to objectively measure individual differences in expert performance. In many professions, such as teachers, doctors, and psychotherapists, this is a daunting task. Traditionally these types of experts have been subjectively evaluated by students, clients, and/or colleagues. But knowing how well an expert is liked or admired does

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not necessarily tell us whether that expert is more able than other experts. More recent research has tested students' performance at the beginning and end of the school year. By comparing the average student's improvement under each teacher, we can identify those teachers who induce bigger increases in their students' performance. Interestingly, we have learned that teachers' effectiveness does *not* correlate with their own professional training. The correlation with length of experience is also quite limited.[2] The same lessons apply to measuring effectiveness of psychotherapists,[3] and experts in other domains.[4]

Conversely, many domains of expertise have a long history of successfully measuring performance. In sports, for example, venues allow athletes to perform under fair and equal conditions; one can therefore measure the time to complete a certain distance in running, bicycling, and swimming, or how high/far someone can jump. Organized competitions allow gymnasts, divers, ice skaters, and dancers to perform their practiced routines in front of trained judges.

In the middle of the 20th century, Elo[5] developed a system of chess ratings based on a mathematical analysis of the outcomes of matches in chess tournaments. These ratings are highly predictive of outcomes of matches between two rated chess players. Chess ratings provide objective data on the current level of skill attained by players as well as longitudinal data on their performance across the life span. More recently, worldwide competitions have emerged in other domains such as music, mathematics, chemistry, and memory. In the memory world championship, competitors from many countries gather to compete in a number of events, such as recall of spoken random digits and speed of memorizing a deck of cards.

Such objective measures are an important first step in developing a coherent theory of expert performance—but only a first step.

THE STUDY OF SUPERIOR EXPERT PERFORMANCE

The point has already been made that extended training time is not sufficient to attain world-class expertise. It is, however, necessary. In a seminal paper, Simon and Chase[6] proposed that one needs approximately 10 years of full-time engagement to become an expert. Indeed, it is unprecedented for a performer in any domain to reach a national or international level of expertise without many years of engagement. Most experts are recognized sometime during their 30s or 40s. These are the ages when scientists are most likely to make major scientific contributions, and when authors and composers are most likely to create their most valued artistic contributions.[7] In sports, peak accomplishments tend to occur earlier; athletes typically break world records in their 20s and 30s. To reach the level of an international chess master, a player must have intensively played chess for over 10 years.[6]

Simon and Chase proposed the first general theory for the acquisition of expertise through pattern-recognition. When first introduced to a new domain of activity, they argued, individuals generate their actions 'by slow, conscious heuristic search'. Then, after extended practice, they see patterns 'quickly and unconsciously' and the appropriate actions come to mind 'almost automatically' (Ref 6, p. 403). This theoretical analysis of expertise led Chase and Simon[6] to search for the acquired patterns by asking chess players to reproduce a briefly presented chess position. They found that a chess master could typically reproduce most of the pieces of chess positions with over 20 pieces, whereas a beginning chess player could only recall an average of 4 pieces. Most importantly, when chessboards with the same chess pieces were randomly rearranged before being presented, the elite chess players' recall was reduced virtually to the level of the beginners, recalling only 4–6 pieces. Subsequent research has confirmed that experts *do not have superior general memory abilities*. Rather, experts' superior memory abilities are restricted to meaningful stimuli from their domain of expertise and *do not generalize* to other types of information.[8] The Simon-Chase theory inspired a large body of research comparing the performance of experts and novices.[9]

A different approach to understanding the development of expertise was taken by Dreyfus and Dreyfus,[10] who collected and analyzed 'descriptions of changes in the perception of the task environment reported by performers in the course of acquiring complex skills' (p. 1). Although their approach was influential,[11] a review of studies of actual performance of nurses found no correlation between amount/level of nurse experience and objective performance.[12] Taken together with findings from other domains of expertise, there is a clear lack of association between length of experience and performance, and between *perceived expertise* and performance. The consensus of researchers in the field is that we must focus on the reproducibly superior performance of experts, not the behavior of individuals.

Along with Simon and Chase, the pioneer for studying expertise was de Groot,[13] the first person to design experimental tasks that could capture the superior performance of experts under controlled conditions. de Groot[13] selected complex positions

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from actual chess games and asked chess players of different skill levels to select the best move for these positions while 'thinking aloud'. He found that the best players were much better able to find the best move than less skilled players. He then analyzed the differences in thought processes to identify the hypothetical causes of the superior performance.

de Groot's[13] research inspired the *expert-performance approach* to the study of expertise.[14,15] The first step in the expert-performance approach (see Figure 1, top portion) is to identify representative tasks and reproduce the experts' consistent superior performance in a controlled laboratory situation. A good example is de Groot's[13] use of the best-move task. Similarly, one can have athletes or other individuals generate actions in a simulator or in response to a video sequence.

In the second step, investigators record observable information on the experts' and less-skilled participants' processes by collecting 'think aloud' protocols and patterns of eye-fixations. For example, an elite tennis player can anticipate (better than chance) where a tennis ball will land before the server has made ball contact; this implies an ability to anticipate based on cues in the server's preparatory movements.[17] Initially, experts were thought to have an amazingly fast neural system, but careful experimentation revealed superiority linked to anticipation based on early movement cues from the server. *In fact, the speed of reacting to a simple stimulus, such as a light or sound, with a key press is not faster for athletes than for non-athletes.*

Another observed skill is experts' ability to plan, monitor, and evaluate their performance. Evidence from many domains shows that experts can reason about hypothetical situations in their domain of expertise. For example, chess masters can, without special training, play chess without seeing a chessboard ('blindfold chess'). Blindfolded players verbalize their thoughts about alternative sequences of moves and play almost as well as when not blindfolded. But note: these chess masters have not demonstrated any superior memory outside of chess!

The expert advantage has been found to be domain-specific and does not generalize to other domains and altered activities. For example, only a small number of tennis players are left-handed. Consequently, both right-handed and left-handed tennis players get more training playing against right-handed competitors. This mismatch in relevant practice experience gives left-handed players a clear advantage over right-handed players in tournaments.[18]

The high degree of specificity of mechanisms mediating expert performance—for instance, differential practice and skill in planning, monitoring, and evaluating performance—suggests that these mechanisms must be acquired. Further, as illustrated in Figure 2, research suggests that they are acquired by gradual increases in complexity over extended time.

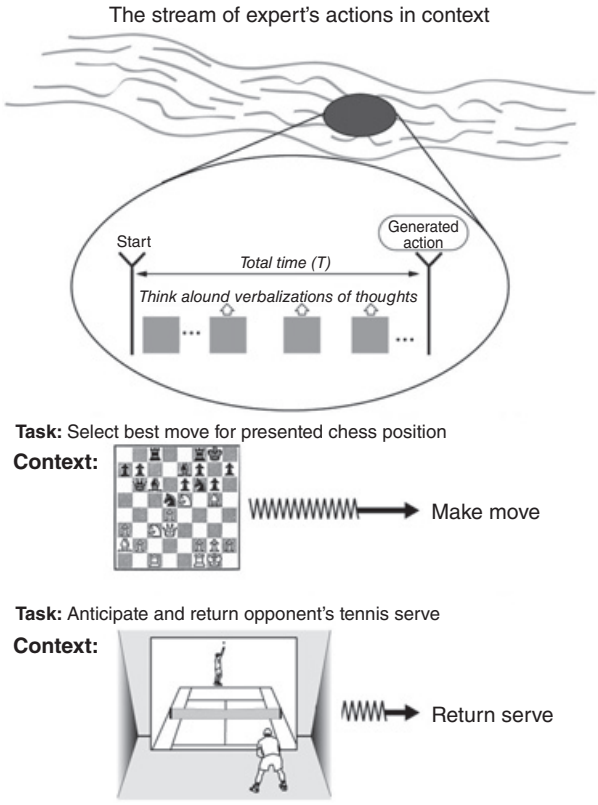


FIGURE 1 | Procedure for capturing expert performance in the laboratory by representing naturally occurring situations as the task. The top of the figure shows the stream of naturally occurring situations. As shown in the bottom part of the figure, one can present a particular challenging chess position from an actual game and ask for the best next move in a laboratory setting. Alternatively, one can project a film of a player serving a tennis ball and ask the participants to attempt to return the 'imaged ball' with the best shot using their racket. These representative situations can then be presented to participants differing in skill, and their performance can be recorded with reaction times, accuracy of response, and process-tracing data—such as eye-fixations and concurrent or retrospective verbal reports. (Re-publication of Figure 1 in Ericsson, K. A., & Ward, P. (2007). *Current Directions in Psychological Science, 16,* 347)

DELIBERATE PRACTICE

In many domains of expertise, longer periods of experience do not by themselves lead to higher levels of performance. Simply doing the same thing again and again does not increase the quality of performance; it merely makes it less effortful and more automatic. So what does lead to better performance?

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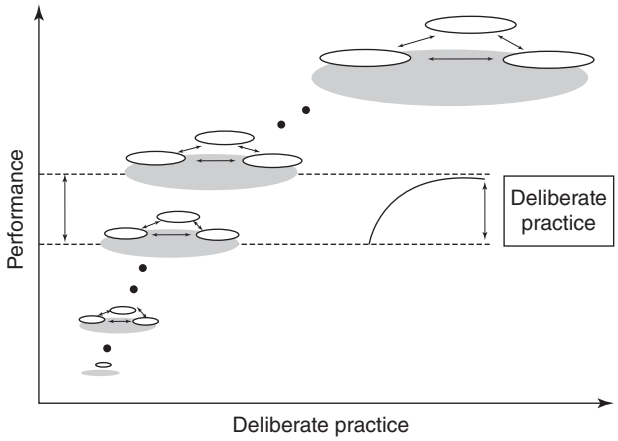


FIGURE 2 | A schematic illustration of the acquisition of expert performance as a series of states with mechanisms of increasing complexity for monitoring and guiding future improvements of specific aspects of performance. (Adapted from "The scientific study of expert levels of performance can guide training for producing superior achievement in creative domains" by K. A. Ericsson in *Proceedings from International conference on the cultivation and education of creativity and innovation* (p. 14). Beijing, China: Chinese Academy of Sciences. Copyright 2009 International Research Association for Talent Development and Excellence)

To find the type of effective learning that has reliably produced exceptionally high performance for centuries, Ericsson et al.[20] turned to the domain of music. They found that the key to improvement is 'deliberate practice', which is engaging in practice activities with a clear goal of improving a specific aspect of performance beyond its current level. These training activities require accurate and detailed *immediate feedback* with opportunities for repetitions after problem solving and/or reflection in order to *gradually* attain the desired performance. The training goals and activities need to be appropriate for the individuals so they can attain the target goal within a limited time with their current skill level—a judgment that typically requires an experienced teacher or coach. For each musical instrument, there is a curriculum with increasingly difficult skills that guides this development.

How much training is required to reach the highest levels of violin performance? To answer that question, Ericsson et al.[20] recorded the practice of students at an international music academy in Berlin. All students had weekly meetings with their master teachers to identify targets for improvement, and were assigned training tasks to improve particular aspects of their performance. The students would then practice by themselves. Ericsson et al.[20] collected estimates for the number of weekly hours across their career and calculated the accumulated number of hours of solitary music study for each year of each musician's development. By age 20, the top violinists had accumulated on average 10,000 h of practice with a standard deviation of around 2000 h. The average of this top group was roughly 2500–5000 h larger than the averages of two other groups of less accomplished violinists in the academy. Thus, greater amounts of accumulated solitary study by each musician was correlated with higher music performance—rejecting the prevailing expectation that more 'gifted' musicians needed less practice.

In the last 20 years, there has been a search for training activities that meet the criteria for deliberate practice. For example, when people play chess, they do not get immediate feedback when they make an inferior move and might not detect the problem until the end of the game. This lack of immediate feedback helps explain why the number of hours playing chess does not predict a player's ranking. A much more effective practice activity involves studying the games of chess masters. During this activity, the player can pretend to play against a world-class player by trying to generate the best move for each chess position. Once the player has planned out different move sequences and selected the best move, she can immediately compare the generated move to a chess master's published move. Then the player can try to determine why her planning did not discover the merits of that particular move. This process provides the chess player with immediate feedback for each move—meeting the criteria for deliberate practice. Charness et al.[21] found that the number of hours that players reported having spent engaged in solitary practice during their careers was highly correlated with their chess rating. The deep exploration of various chess move sequences while selecting a move also enhances the development of memory skills for generating and recalling moves, which can explain chess masters' ability to play blindfold chess without additional practice.

In medical practice, doctors often discover errors in their diagnoses months or years after the patient finally recovers. This delayed feedback can do very little to enhance their skill. A deliberate practice method would have doctors and medical students diagnose X-rays and other types of images for patients from old cases, where the correct diagnosis is known. This would allow for immediate feedback. In recent decades, sophisticated simulators have allowed medical students to practice medical procedures without harming patients. Simulator experience in itself, however, is not sufficient for improving performance. Issenberg et al.[22] showed that for simulator training to reliably improve performance, it is

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necessary to specify a particular performance goal of speed and accuracy, and then have the students perform the procedure repeatedly with immediate feedback until they reach the required goal in a reproducible manner.

Whenever training that meets the criteria for deliberate practice has been developed, it has been found to be associated with increases in performance in numerous domains of expertise, such as nursing, ballet, sports, SCRABBLE, and scientific research.[14]

THE ROLE OF BIOLOGY

There is general agreement that training, including deliberate practice, is necessary to attain expert performance. But what is the role of individual biological differences? Although it is widely observed that the bodies of many types of athletes, including weightlifters and basketball players, are distinct from their non-athletic peers, it does not follow that these differences are inborn and inevitable. Extended intense training where athletes push themselves beyond their comfort zone is known to stimulate the body to initiate physiological and anatomical changes. In particular, it spurs muscular development and blood supply. In addition, there is clear evidence that intense training at relatively young ages is necessary for modifying joints to make them suitable for ballet and baseball pitching.[23]

The brain is also powerfully influenced and shaped by training. In recent years, neuroplasticity has been shown to be integral to adult learning and development[24] (see Power and Schlaggar, Neural plasticity across the lifespan, *WIREs Dev Biol*, also in the collection How We Develop). A recent review[25] found systematic differences in the structure of the brains of adult experts in basketball, golf, judo, and diving compared to the general population, where the differences were most likely the result of extended practice. The links between neuroplasticity and extended training are particularly compelling in a recent study that examined black belt Karate experts producing punches with maximal force[26]; it was found that the performance characteristics of each individual's punch were related to acquired differences in the white matter of the cerebellum, a brain structure that is important for motor control. Such training-induced effects on brain structure are becoming increasingly evident.[27]

With our modern understanding of gene–environment interaction, gene expression, and epigenetics, we can no longer seriously entertain the age-old concept that we possess inborn talents or traits that develop independently of experience and training[23] (see Charney, Genes, behavior, and behavior genetics, *WIREs Cogn Sci*, and Moore, Behavioral epigenetics, *WIREs Syst Biol Med*, also in the collection How We Develop). Indeed, virtually every known physical and mental characteristic related to expert performance is modifiable.[28] Accordingly, the more realistic view is that individuals' performance in any domain is influenced by complex interactions among experience, training, and biological factors during childhood, adolescence, and adulthood. More research is needed to identify the optimal sequence of training activities for an individual—starting in early childhood—to discover the practical limits of her attainable performance.

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